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Chapter 5: Naming

Objectives

At the end of this Chapter Students be able to know:-

- Naming Entities
- Flat Naming
- Structured Naming
- Attribute based naming

Introduction

- names play an important role to:
 - share resources
 - uniquely identify entities
 - refer to locations etc.
- an important issue is that a name can be resolved to the entity it refers to.
- to resolve names, it is necessary to implement a naming system.
- in a distributed system, the implementation of a naming system is itself often distributed, unlike in non-distributed systems
- efficiency and scalability of the naming system are the main issues

Naming: Names, Identifiers, and Addresses

➤ NAME

- A name in a distributed system is a string of bits or characters that is used to refer to an entity.
- E.g. of entity,
 - resources such as hosts, printers, disks, and files
 - well-known examples of entities that are often explicitly named are processes, users, newsgroups, Web pages, messages, network connections, and so on.

Why Name?

- Convenience in Accessing/Sharing Objects (Files)
- Fundamental concept in Communication

Naming: Names, Identifiers, and Addresses

- Entities can be operated on. For example, a resource such as a printer offers an interface containing operations for printing a document, requesting the status of a print job, and the like.
- **Addresses**
 - Identifiers of places to find the named entities.
 - a special kind of name: it refers to an **access point** of an entity.
 - The name of an access point is called an address(such as IP address and port number as used by the transport layer)

Naming: Names, Identifiers, and Addresses

➤ Identifiers

- names that are used to uniquely identify an entity.
- Names in DS are most commonly true identifiers.
- A true identifier is a name that has the following properties :
 1. An identifier refers to at most one entity.
 2. Each entity is referred to by at most one identifier.
 3. An identifier always refers to the same entity (i.e., it is never reused).

Example — a Person

- Name — Solomon Ayele
- Address — Kebele 03 ,Gubre, Wolkite
- Route —
 - From main road, go west 1 km
 - Turn right, then take 3rd left
 - He lives at the 2nd house on the right

Example — a File

- Name — **index.html**
- Address — inode **#54321**
- Route (path) — **/csopt1/cs4513/public_html/d07/index.html**

Naming System

- In many computer systems, addresses and identifiers are represented in machine-readable form **only**.
- In contrast to addresses and identifiers, a human-friendly name is generally represented as a character string. These names appear in many different forms.
 - how do we resolve names and identifiers to addresses?
- Name resolution maps a **name** to its **address**.
- Name resolution is performed by a naming system
- a **naming system** maintains a name-to-address binding which in its simplest form is just a table of (name, address) pairs.

Naming System



- A naming system is simply a middleware that assists in name resolution

Types of Naming systems

- Flat naming
- Structured naming
- Attribute-based Naming

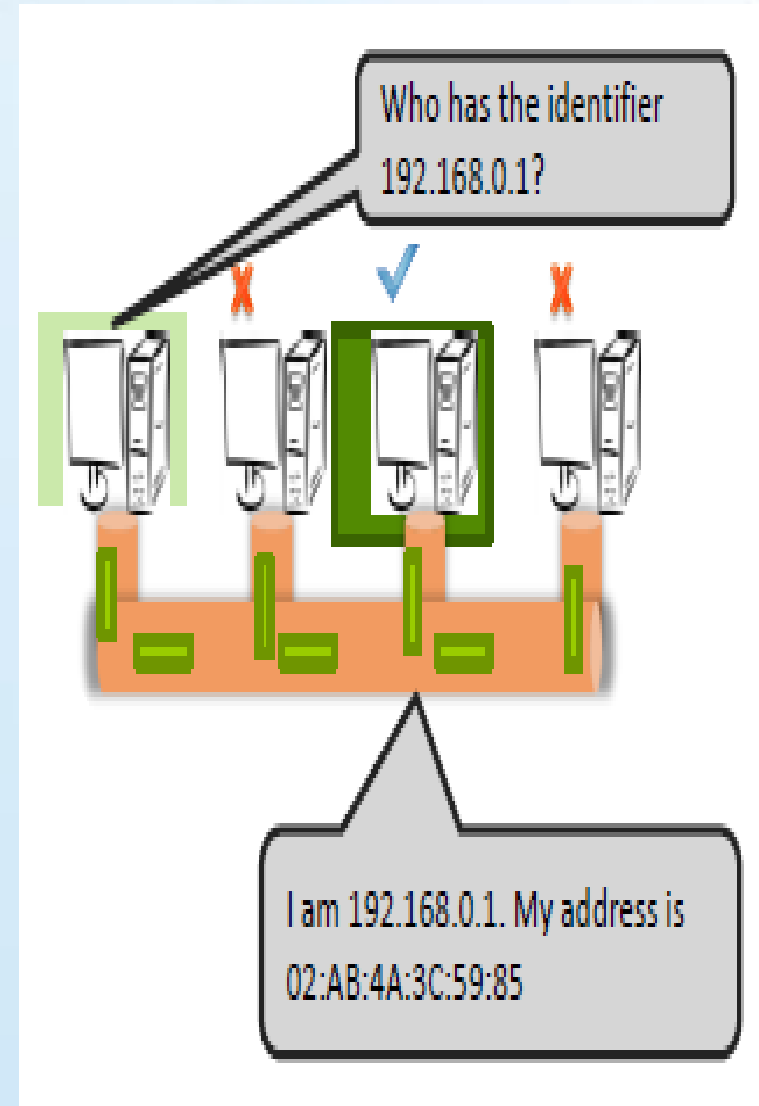
Naming Systems(contd.)

Flat Naming:

- In a flat naming system, the identifier or name assigned to an entity is a random bit string or a unique identifier without any inherent meaning or structure.
- The name itself does not provide any information about how to locate the associated entity or any characteristics of the entity.
- This type of naming system is often used in machine-to-machine communication or internal systems where the primary goal is to ensure uniqueness rather than human comprehension.

Naming Systems(How to resolve flat names?)

- **Broadcasting and Multicasting**
- Broadcast the identifier to the complete network.
- The entity associated with the identifier responds with its current address.
- Resolve an IP address to a MAC address.
- IP address is the identifier of the entity
- MAC address is the address of the access point



Naming Systems(How to resolve flat names?)

- Forwarding Pointers

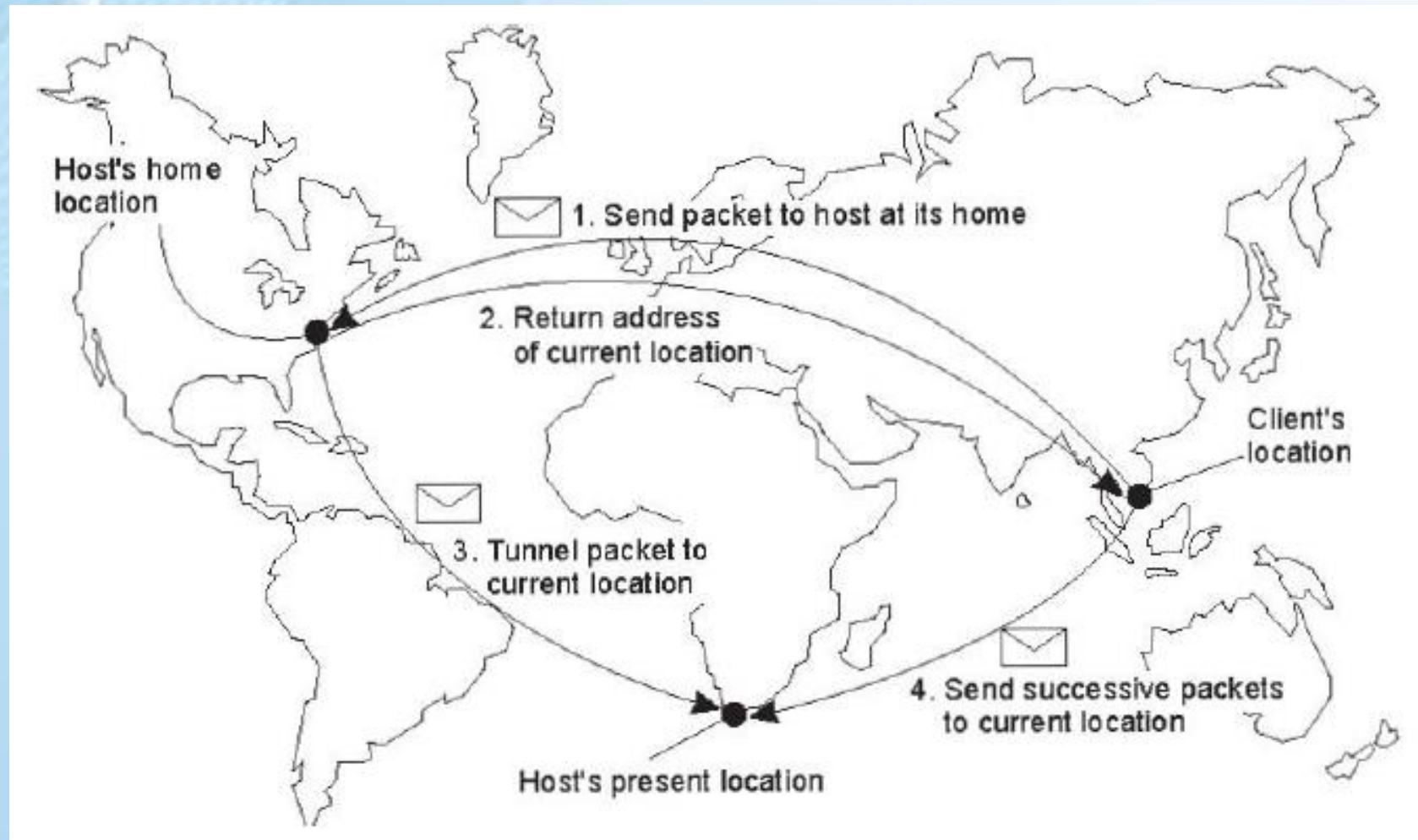
- it enables locating mobile entities.
- Mobile entities move from one access point to another.
- When an entity moves from location A to location B, it leaves behind(in A) a reference to its new location at B
 - Name resolution mechanism
- Follow the chain of pointers to reach the entity.
- Update the entity's reference when the present location is found.

Naming Systems(How to resolve flat names?)

- Home-based approaches

- Each mobile entity has a fixed IP address(home address)
- Entity's home address is registered at a naming service
- Home node keeps track of current address of the entity(care-of address)
- Entity updates the home about its current address (care-of address) whenever it moves
 - Name resolution mechanism
- Client contacts the home to obtain the current address
- Client then contacts the entity at the current location

Naming Systems(How to resolve flat names: Home-based approaches: Example)



Structured Naming

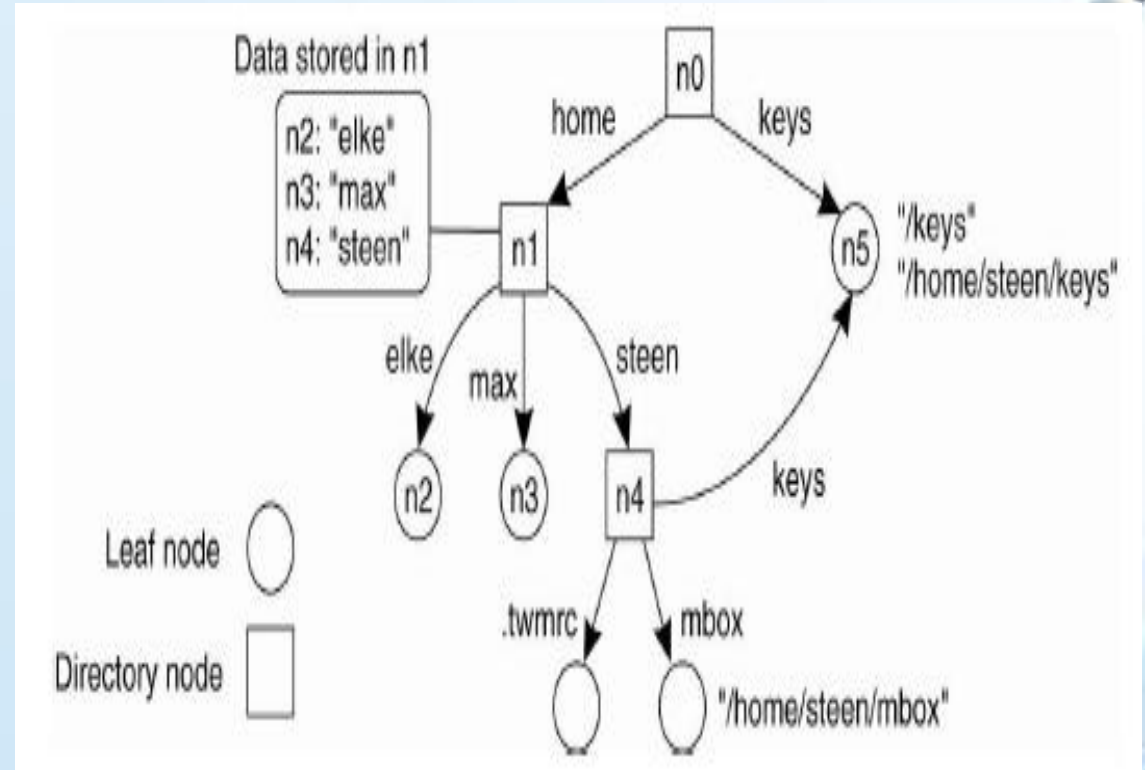
- Flat names are difficult for humans to remember
- Human-friendly names are structured names composed of several parts
- Example: filenames, hostnames
 - /home/userid/work/dist-systems/naming.txt
 - www.wku.edu.et
- How to locate an entity given its human-friendly name?

Name Resolution

- The process of looking up a name is called **Name Resolution**.
- Resolving a name in a name space results in a reference to the node referred to by that name.
- Name resolution is performed by traversing the graph
- Knowing how and where to start name resolution is referred to as a closure mechanism.
 - Keep a reference to the root node for dealing with **absolute path names**.
 - Keep a reference to the **current** directory node for dealing with **relative path names**

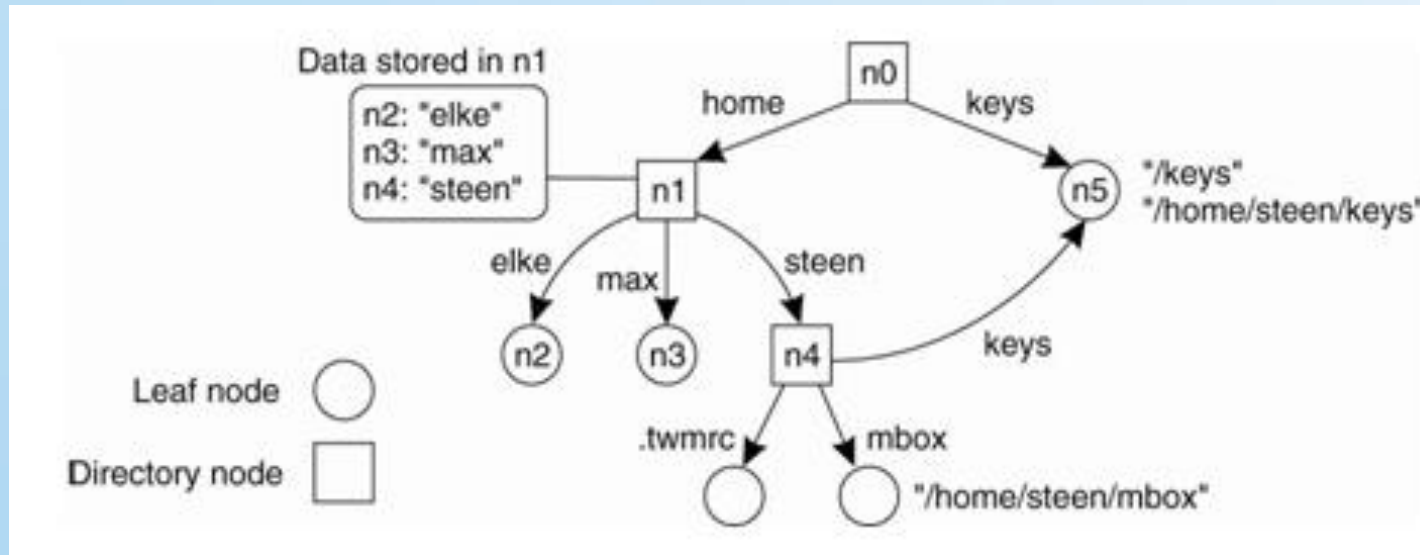
Structured Naming: Name Spaces

- Structured Names are organized into name spaces
- A structured name space is represented as a labeled directed graph with two types of nodes
- A leaf node represents a named entity and stores information about the entity
- A directory node stores a directory table of (edge label, node ID) pairs



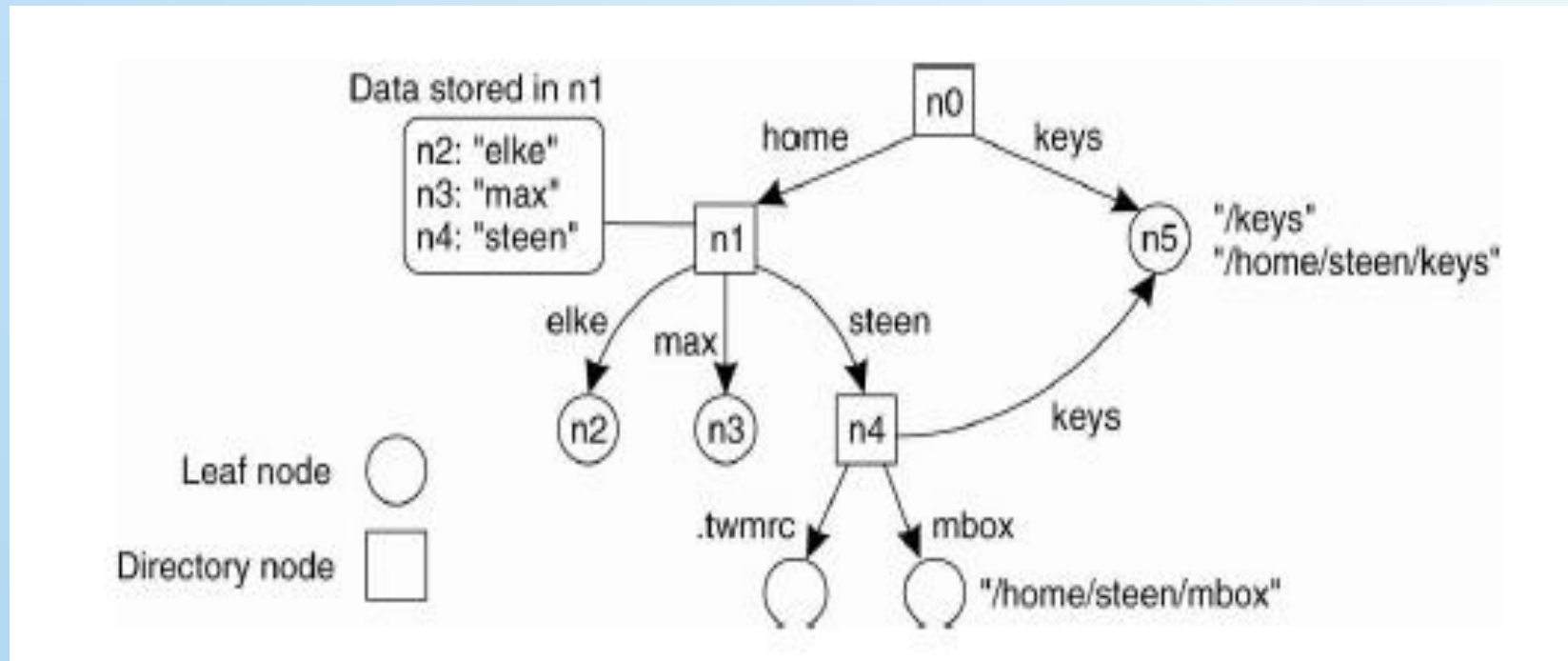
Structured Naming: Name Spaces

- A path name is used to refer to a node in the graph
- A path name is a sequence of edge labels leading from one node to another.
- An **absolute path** name starts from the root node (e.g., /home/steen/mbox)
- A **relative path** name does not start at the root node (e.g., steen/mbox)



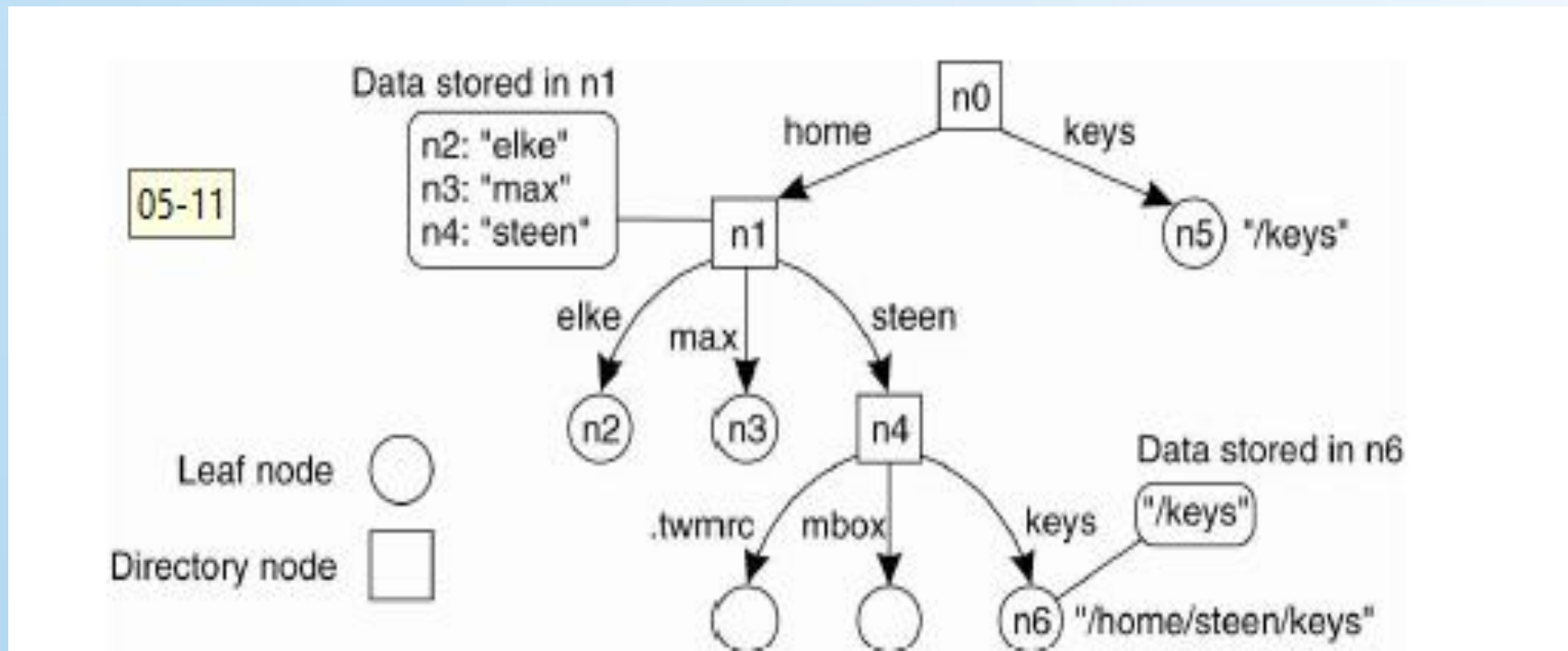
Structured Naming: Aliasing

- **Aliasing**: an entity may be reachable by multiple paths and therefore is named by multiple path names.
- When multiple path names refer to the same node, these path names are called hard links to the node. E.g., path names **/keys** and **/home/steen/keys** are hard links to node n5 (see following graph)



Structured Naming: Aliasing

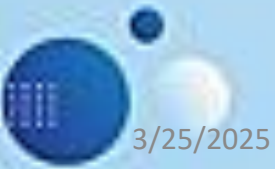
- A symbolic link occurs when a leaf node holds a path name that refers to another node
- E.g., the path name /home/steen/keys is a symbolic link to node n5 (see graph below)



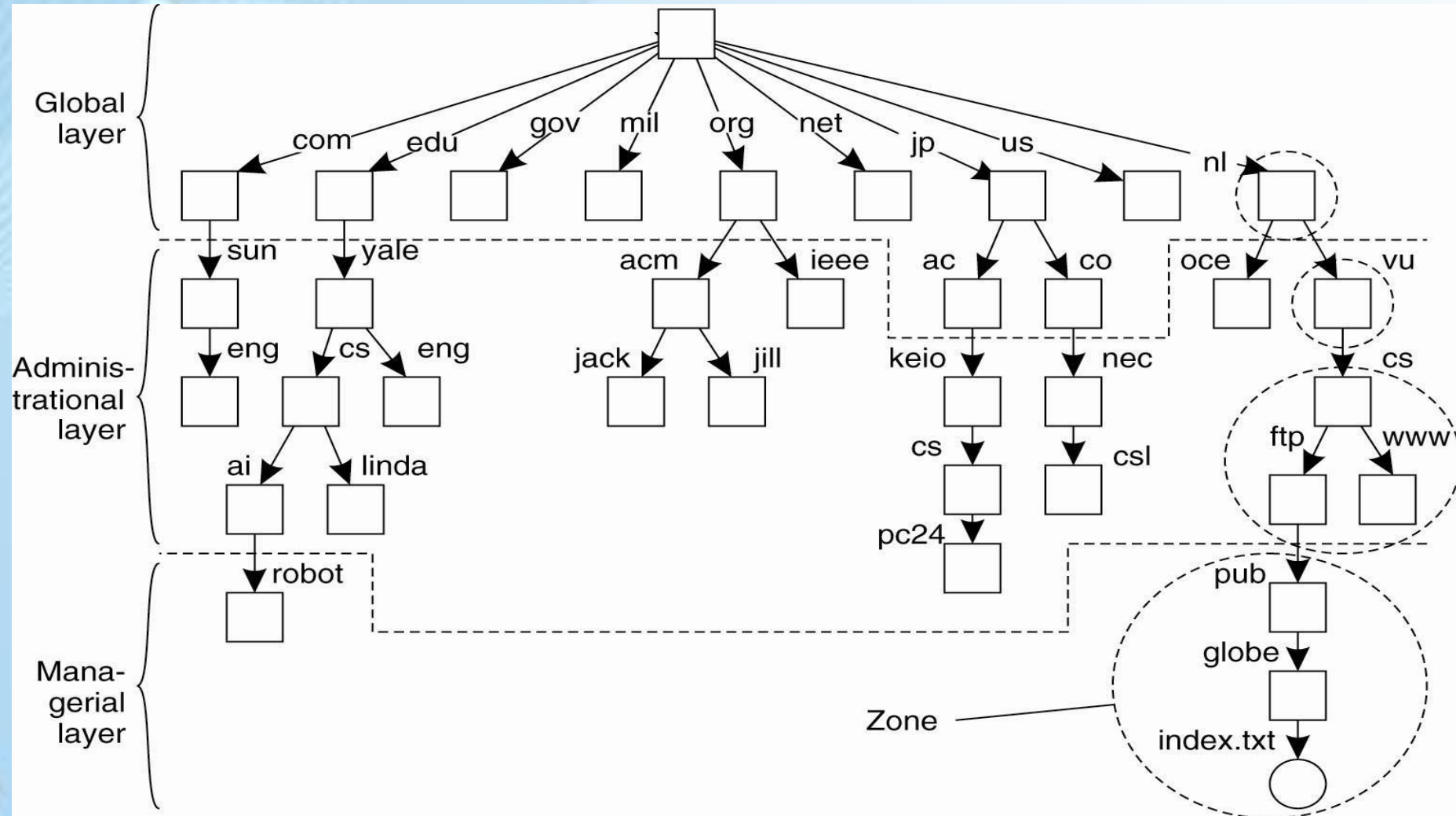
Name Space Implementation



- Name space for a large-scale distributed system is partitioned into three logical layers.
- **Global layer:** Generic domain name
 - high-level directory nodes(root and its Children)
 - Represent groups of organizations, directory tables rarely changed.
- **Administrational layer:** directory nodes managed by a single organization.
 - Typically one directory node per department, directory tables infrequently changed
- **Managerial layer:** nodes that change frequently.
 - Example: nodes representing hosts, user-defined directories and files



Name Space Implementation(contd.)



Attribute-based naming



- Attribute-based naming allows an entity to be described using (attribute, value) pairs, where the attribute represents a characteristic or property of the entity, and the value represents the specific value or information associated with that attribute.
- As more information becomes available, efficient searching for entities becomes crucial.
- Users should be able to search based on specific attributes to find relevant entities.
- Each entity has certain attributes that provide information about it.
- Attributes describe different aspects or characteristics of the entity.

Attribute-based naming



- Users can search for entities by constraining the attributes. This means they can specify certain attribute values or ranges to filter their search results and retrieve entities that meet their criteria.
- Attribute-based naming systems are also known as directory services(Directory services are used to organize, manage, and retrieve information about entities based on their attributes).
- These systems organize entities based on their attributes, making it easier to search, locate, and manage them.

Attribute-based naming

- The Resource Description Framework (RDF) is a more general model used for describing resources.
- Resources are represented as triplets consisting of a subject, predicate, and object. For example, (person, name, Alice) represents the fact that the subject "person" has the attribute "name" with the value "Alice."

Thank You!!

